Clinical Guideline for:
High Frequency Oscillatory Ventilation (HFOV) in Neonates

Summary

This guideline outlines the use of high frequency oscillatory ventilation in neonates.

Criteria for consideration of commencing on HFOV

- Difficulty with Oxygenation despite optimising conventional ventilation.

  or

- Difficulty with CO_2 clearance despite optimising conventional ventilation.

A senior paediatrician must always be consulted prior to commencing HFOV.

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1. INTRODUCTION

HFOV is reserved for infants with moderate to severe RDS as early intervention and rescue therapy. Evidence of benefit over conventional ventilation (CV) is weak with Cochrane review concluding that *"there is no clear evidence that elective HFOV, as compared with CV, offers advantages when used as the initial ventilation strategy to treat preterm babies with acute pulmonary dysfunction. There may be a small reduction in the rate of CLD with HFOV use but the evidence is not significant".*

However, clinical experience shows some individual babies benefit from HFOV therefore its use may be considered with the following criteria: HFOV may be effective in other conditions such as persistent pulmonary hypertension of the newborn, meconium aspiration syndrome and pneumothorax.

2. PRINCIPLES OF HFOV

- Mean airway pressure (MAP) and FiO_2 determine oxygenation.
- Oscillation amplitude (Δp) and frequency (Hz) affect tidal volumes and CO_2 clearance. CO_2 clearance can be monitored by DCO_2.

3. CRITERIA FOR CONSIDERATION OF COMMENCING ON HFOV

- Difficulty with Oxygenation despite optimising conventional ventilation.

  or

- Difficulty with CO_2 clearance despite optimising conventional ventilation.

A senior paediatrician must always be consulted prior to commencing HFOV.

4. CHOICE OF VENTILATOR

We have 4 Drager VN500 ventilators that have a good capacity for oscillation.

5. INITIAL HFOV SETTINGS

Instructions for changing the VN500 from Conventional ventilation to HFOV can be found attached to ventilator.

i. Ideally invasive blood pressure monitoring should be in situ, if not record 15 minute 'Dinamap' BP whilst optimising ventilation.

ii. Starting frequency (Hz) should start at 12Hz for babies <2kg and 10Hz for babies >2kg and should not be changed unless CO_2 retention is unresponsive to other manoeuvres.

iii. Commence at a MAP 2cm H_2O greater than required MAP that has been used on CV.

iv. Choose an amplitude of oscillation (ΔP or pressure) that results in good chest wall movement (usually at least MAP + 10cm H_2O)

6. OXYGENATION

- If lung inflation is adequate, SaO_2 should improve and the FiO_2 can be reduced.
- If the infant remains hypoxic, continue to increase MAP cautiously every 5-10 minutes until FiO_2 can be reduced to 30%.

  (Remember that if the lungs are over inflated SaO_2 will improve with a lower MAP)
- Lung compliance will improve rapidly after surfactant and MAP will need to be reduced to avoid over-inflation.
- If in doubt always reduce MAP by 1-2 cm water before increasing.

Over distension with too high MAP causes poor compliance, poor venous return, poor oxygenation and poor CO_2 clearance. This can be checked with CXR.

7. CO_2 CLEARANCE

If high CO_2 remains a problem, increase ΔP (delta). Oscillations of higher amplitudes facilitate CO_2 clearance by increasing tidal volume. Unlike CMV, it is rarely necessary to alter the frequency. Any changes to the frequency should be discussed with an experienced neonatal paediatrician.

Decreasing frequency increases CO_2 clearance.

If the infant is being ventilated effectively, there should be quiet breathing. The infant should be neither recessing nor apnoeic. If either occurs, the ΔP may need adjusting to control CO_2.

8. VOLUME GUARANTEE WITH HFOV

This can be set to achieve a more stable $PaCO_2$. If VG is set for HFOV then reducing frequency will REDUCE PaCO2 clearance (i.e. increase CO2 levels) and increasing frequency will INCREASE PaCO2 clearance (i.e. decrease CO2 levels). Set VG at approximately 1.5-2.5mls/kg. The ventilator will adjust Δp to achieve targeted volume.

CO2 clearance is determined by DCO2 which = VT^2 x f

9. SUCTION

Avoid disconnection if at all possible. You will lose lung volume very rapidly once you disconnect and it may take some time to re-recruit collapsed alveoli and achieve oxygenation again. (You may need to increase MAP for a while after disconnecting until lung volume stabilises).

Suction should be performed using *inline suction catheters*. Please refer to separate guidelines for suctioning procedure.

10. MONITORING LUNG INFLATION

Frequent CXRs are needed to review lung inflation, checking for sufficient recruitment and avoiding over inflation. If FiO2 is below 30%, progress of the infant can be monitored using arterial blood gas monitoring without frequent CXRs.

Tachycardia and/or reduced mean arterial blood pressure on invasive monitoring may indicate over distension.

Adequate inflation is deemed to be present when 8-10 ribs can be seen posteriorly.

11. WEANING FROM HFOV

This can be done by cautious reduction of the MAP and ΔP. It is usual to transfer back to conventional ventilation prior to extubation when the MAP is <8 cm H_20 and $Fi0_2$ is <30% but it is also possible to extubate straight to CPAP from HFOV.

12. SEDATION

The infant should be sedated with morphine but doesn't necessarily need tobe paralysed.

13. NURSING CARE AND OBSERVATION OF INFANTS ON HFOV

- Standard nursing care for ventilated infants.
- Continuous heart rate & SaO_2 monitoring should be documented hourly.
- At least hourly 'Dinamap' BP if invasive monitoring is not available.
- Respiratory rate monitoring may be turned off during HFOV.
- Observe for chest vibrations.
- Observe for spontaneous respiratory effort. Quiet breathing should be seen over the oscillations. Record respiratory effort hourly on observation chart e.g. none, quiet breathing, recession, excess work of breathing.

14. DRAGER VN500 LEAK MODES

The Drager VN500 has 2 leak modes:

Leak Compensation
This is to compensate for leak around the ETT when delivering tidal volume. The Ventilator assumes that there is leak during inspiration and during expiration. The difference between inspiratory volume delivered and expiratory volume delivered is close to leak percentage however is not exact. The ventilator then makes a calculation on this to estimate the approximate volume delivered to the lungs. This means the volume delivered is closer to target than without compensation.

Leak adaptation
Leak adaptation adjusts so that when there is a leak around an ETT this is not displayed on the screen. This prevents autotriggering as the expiratory level (with expiratory leak) which would be above the trigger level, does not trigger the breath.

15. REFERENCES

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- Henderson-Smart DJ, Bhuta T, Cools F, Offringa M. Elective high frequency oscillatory ventilation versus conventional ventilation for acute pulmonary dysfunction in preterm infants. The Cochrane Database of Systematic Reviews 2003, Issue 4. Art. No.: CD000104. DOI:10.1002/14651858.CD000104

16. PUBLICATION DETAILS

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